

control systems. The current trend is to use field-programmable gate arrays (FPGAs) and system-on-chip (SoC) or system-on-module (SoM) solutions that combine several components into one small form factor.

Freescall, NXP Semiconductors, Bosch, Infineon, ST Microelectronics and Renesas are amongst some of the key suppliers to the auto industry, offering a horde of solutions for sensing, in-vehicle networking, automotive lighting, car radio and audio to automotive power and telematics on-board units to access and immobilisation. Most of these companies also offer associated design tools.

More updates.

► Nvidia announced the Tegra K1 processor at CES 2014. With 192 cores, it claims to be the hardware that can fully enable the future autonomous car. Audi will be using the K1 chip in production cars in the near future.

► Bosch offers single-chip airbag system ICs that combine power supply, sensor interfaces, I/O functions, firing loop drivers and safing engine on one single chip. The semiconductor industry offers several SoCs, such as angular rate sensors for vehicle dynamics control (VDC), and combined inertial sensors for VDC and active suspension.

► The intensive developments in ADAS and infotainment have led to far greater semiconductor requirements, which the semiconductor world is quickly gearing up to meet. Freescale, for example, offers the iMX6 family, which provides up to four high-performance ARM Cortex-A9 cores, advanced graphics and video accelerators for processing of up to five 1-megapixel video streams at 30 fps for surround view part assist system.

My car, my pal

Most people love their car like it is their family member. So, would they not like to communicate with their car using more natural communication techniques such as speech or gestures? This is nice, not only from a personalisation aspect, but also from the

perspective of safety.

SYNC, an in-car voice-interactive technology featured by Ford helps drivers to pay attention to the road without having to use their hands for making phone calls, browse phone contacts, read texts and browse music. "It helps keep things simple and safe, while on the move," says Piparsania.

Ford also recently announced another technology at the Mobile Developer Summit 2013 in Bangalore—the SYNC AppLink, a technology that allows greater communication between smartphone apps and AppLink-equipped vehicles through voice commands. It will be launched in India in 2014. "This technology will allow greater integration with a range of smartphone apps. AppLink is a Ford-created software technology and the industry's only system that can control smartphone apps via voice control. The platform is planned to be introduced in India with ESPN cricinfo, MapmyIndia, burrrp!, Glympse and TuneIn Radio while encouraging app developers to come up and add new applications," says Piparsania.

"Enhanced gesture control is a strongly emerging trend," says Gupta of Freescale. Audi recently introduced the MMI operating system that enables drivers to control various in-car functions using the MMI interface, touchpad to voice. With this tech, one can easily change and use various controls in the car just by a gesture. "Imagine how easy it is going to be to change your radio station or turn on the AC, all by just a gesture," says Gupta. He also speaks of Visteon's HMye cockpit concept, presented for the first time at CES, which features a unique user interface controlled by eye gaze and head direction data, coupled with image attribute tracking.

More updates.

► In the last quarter of 2013, Google applied for a patent for a technology that helps control your car with hand motions. The system uses a ceiling-mounted depth camera and a laser scanner to understand the user's gestures and hand movements and initiate actions accordingly. For example, you

could swipe near the window to roll down the glass, or wave your hand up to raise the volume on the radio.

My car, my safe haven

While navigation, driver assistance and entertainment are all wonderful, it is the safety and security aspects of today's cars that are most intense. Actually speaking, the telematics systems like car-to-car connectivity and most new features do have a positive impact on safety, except, perhaps the on-board entertainment and features such as hands-free texting, which could distract drivers.

Features introduced by GM in its cars, in association with OnStar, have gained quite some reputation in this segment. Advanced telematics has been used to implement a comprehensive automatic crash notification and security system with features like remote vehicle diagnostics, turn-by-turn navigation, the ability to slow down stolen vehicles and the ability to call a helpline when in danger.

More updates.

► BMW has an interesting feature called run-flat tires, which helps you get to the nearest service centre when your tire is about to go flat. Sensors help predict such a situation, an indicator glows in the dashboard and the tire's special structure allows you to drive around 80 km at 80 km/hr to get help before the boom!

My car, my mechanic

From the tires and door to the powertrain, most parts of the car are fitted with sensors and electronic control units (ECUs) today.

Apart from better control and management of the vehicle, the electronics also help in online diagnostics. "We connect to the heartbeat of the car every day to check how it is doing and with that we are not only talking about diagnostics but also prognostics, or knowing the problem even before it happens. This creates a paradigm shift in the relationship between the customer and the manufacturer, creating a new ecosystem of convenience for the customers," says Maini.